

MySQL - Database

Restaurant Management System

Presented By

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Introduction

The Idea

This project is built as way to learn **how Database Management Systems work**. And it's a demonstration of how you can solve real-life problems using digital products, in this case a fully functioning database. And it can make life easier.

What we will need

- XAMPP
- MySQL Server
- Apache Web Server
- Knowledge of SQL (aka. Structured Query Language)

And with all that, we are ready to **begin our journey** into the database world!

Project Goals

- **Utilize**

Utilizing the **theory** and **MySQL** knowledge to design, develop, and test ideas.

- **Explore**

Exploring fundamentals of **MySQL** in simple practical scenario and utilize it to solve real-world problems .

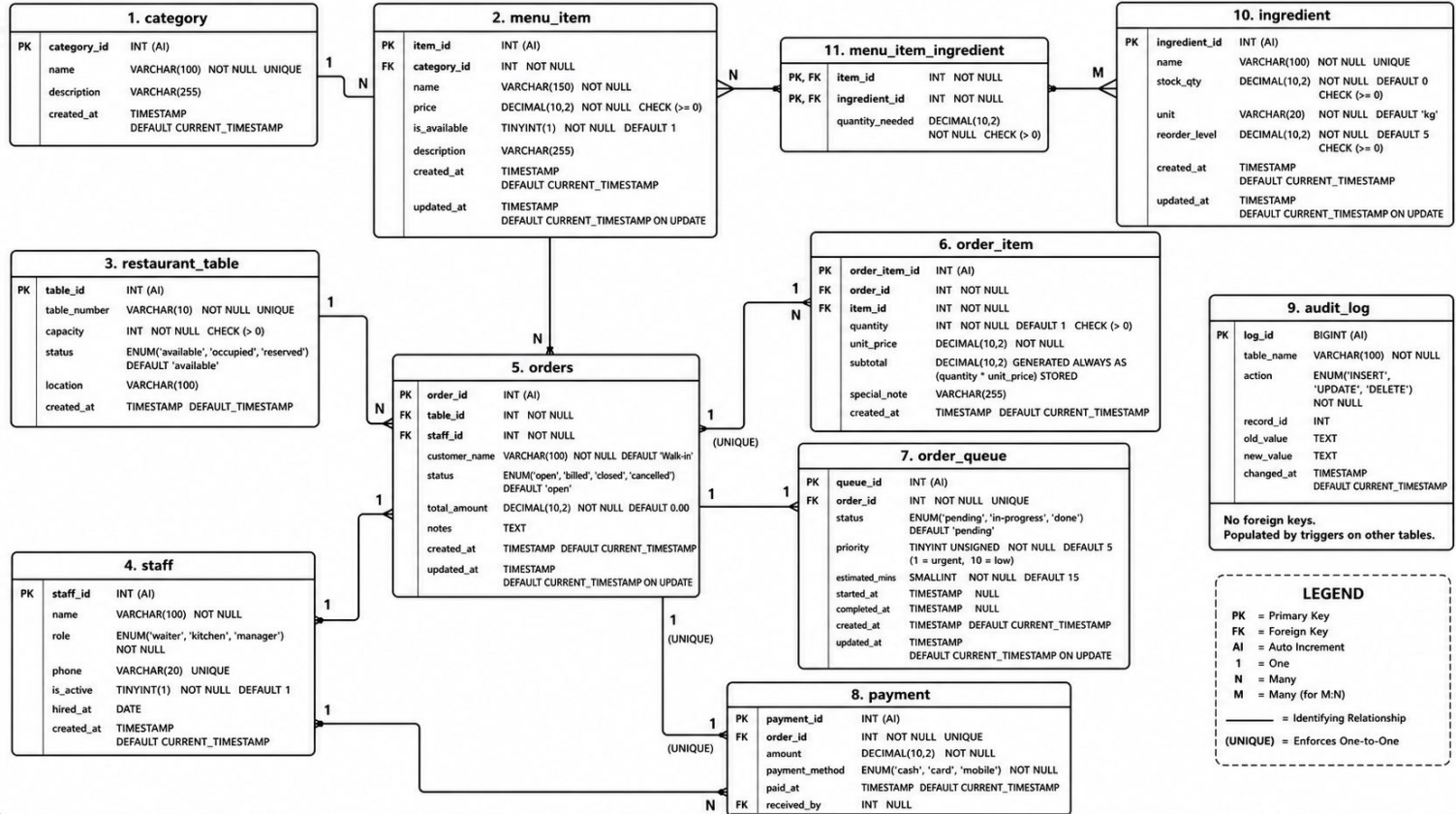
- **Learn**

Learning **DBMS** and **MySQL** through a fun and engaging real-world project task.

- **Output**

Producing clear results that showcase **how to design**, build and finally **utilize** a Database.

ER-Diagram Of the System



Infrastructure Of The System

- **Tables**

- Ingredient
- Menu Item
- Menu Item Ingredient
- Orders
- Order Item
- Category
- Audit Log
- Order Queue
- Payment
- Restaurant Table
- Staff

Infrastructure Of The System

Views

There are total **6** view, that help the manager see the Full menu, Table Status, Active Orders, Daily revenue and etc.

Triggers

There are total **9** triggers, that help to maintain consistency while managing new orders and updating order payment, time, etc.

Procedures

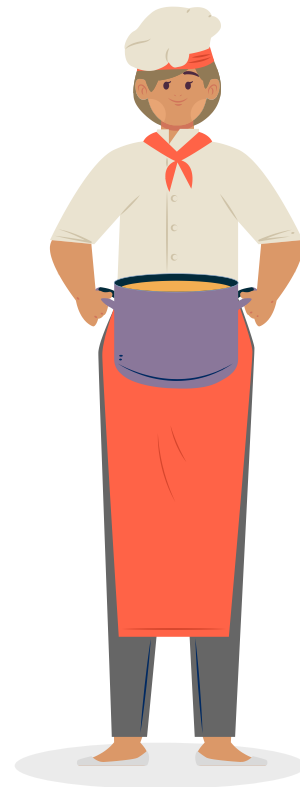
There are total **10** procedures, including one that uses **Cursor** and they help with minimizing the lengthy work like updating queue status, creating new order, etc.

Functions

There are total **5** individual functions, that shortens writing long .

Now let's cook the program

1. Run XAMPP application.
2. Start MySQL and Apache Server.
3. Open <http://localhost/phpmyadmin/index.php>.
4. Run the [restaurant-management-system.sql](#) as raw SQL query.



Future Vision & Conclusion

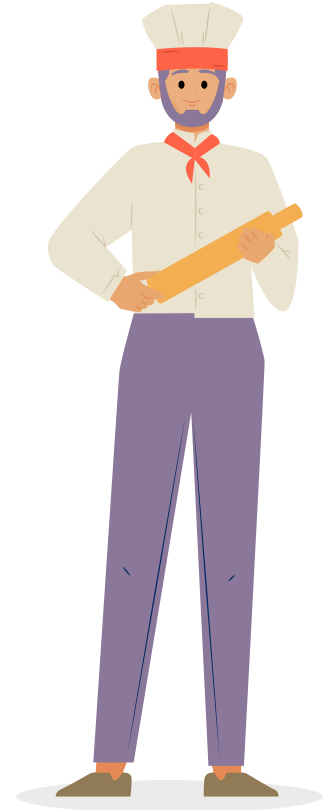
Improvements We can Make

- Add a **Admin management** system.
- Add access security and permission management.
- Introduce **Algorithms** for popular dishes.
- Add **Authentication** and **Billing** system.

Conclusion

This Database delivers a **Normalized** and scalable foundation for restaurant operations. By defining relationship **Cardinality**—from one-to-many order flows to many-to-many ingredient recipes. We ensure strict data integrity and eliminate redundancy.

Coupled with **safeguards** like price snapshotting and an immutable audit log, this design perfectly **balances** daily operational performance with accurate historical tracking.





Thank You!

